



How does government support promote digital economy development in China? The mediating role of regional innovation ecosystem resilience

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ABSTRACT

This study investigates how government support affects digital economy development in China and whether regional innovation ecosystem resilience mediates this relationship. Drawing upon a panel data set containing 31 Chinese provinces from 2011 to 2020, an econometric model is constructed for empirical analysis. We find that government support significantly promotes digital economy development. Regional innovation ecosystem resilience plays a full mediating role between government support and digital economy development. In addition, the spatial spillover effects regional innovation ecosystem resilience on digital economy development is also examined. The results show that regional innovation ecosystem resilience has a positive spatial spillover effect on digital economy development. These findings have important implications for the Chinese policy agenda on digital economy development.

1. Introduction

In recent years, the digital economy has grown rapidly in China. According to the data of the 2019 Digital Economy Report released by the United Nations International Trade, based on the report's definition of the digital economy, the size of the world's digital economy was estimated to be between 4.5 and 15.5 % of the world's gross domestic product (GDP). According to the Ministry of Industry and Information Technology, the size of China's digital economy grew from 11 trillion to over 45 trillion, and China's digital economy of GDP increased from 21.6 % to 39.8 % over the period 2012 to 2021. However, in the first half of 2022, under the impact of the unstable international situation and the epidemic exceeding expectations, China's main economic indicators declined significantly, and the "triple pressure" increased. It has become more difficult to meet the economic and social goals for the year. In this context, maintaining the growth momentum of the digital economy is of great significance to steady growth. Government support is an important measure to deal with downward pressures on the economy, promote high-quality economic development and serve national governance. The Chinese government has issued a series of policies to promote digital economy development. For example, at the overall strategic level, on September 26, 2018, the National Development and Reform

Commission and 19 other departments jointly issued the Guiding Opinions on Developing the Digital Economy to Stabilize and Expand Employment. The aim was to increase employment and ensure stable development of the digital economy. As far as local governments are concerned, on April 19, 2020, the Hebei Provincial Government issued the Hebei Digital Economy Development Plan (2020–2025), which clarified the guiding ideas and implementation path for the development of the digital economy in Hebei Province. This paper therefore aims to explore the relationship of government support and digital economy development.

Extensive research has mainly focused on the measurement of the digital economy (Wang and Shi, 2021; Szeles and Simionescu, 2020; Sidorov and Senchenko, 2020) and its impact on fintech development (Chen et al., 2021), total factor productivity (Pan et al., 2022), and sustainable development (Song et al., 2022). Some studies that have paid attention to the impact of tax reductions (Li and Yang, 2021) and financial investment in science and technology (Zhang et al., 2022) on the development of the digital economy. Only a small number of studies have explored the impact of government-related policies on technological innovation of digital enterprises or industries (Deng et al., 2021; Yu et al., 2021). However, little is known about the link between government support and digital economy development, and two themes are

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often studied separately.

Additionally, regional innovation activities are of great importance for forming competitive advantages and ensuring stable economic development in the context of the digital economy. However, under the “new normal” of the increasingly obvious characteristics of uncertainty, especially the COVID-19 pandemic that started at the beginning of 2020 poses a great threat to the regional innovation ecosystem. For this reason, it is also an urgent problem to improve regional innovation ecosystem resilience, in order to promote digital economy development. But in the particular context of the implementation of the dual-track resource allocation model of market and government in China, the enhancement of regional innovation ecosystem resilience is influenced by government support. Whether government support can promote regional innovation ecosystem resilience, and then favorably influence digital economy development? Previous studies have only explored the impact mechanism of government-related policies on the technological innovation of digital industries or enterprises, which involves enterprise R&D investment (Yu et al., 2021), financing constraints and corporate risk-taking (Deng et al., 2021). However, there is no clear answer on whether regional innovation ecosystem resilience mediates the relationship of government support and digital economy development.

Notably, resilience is defined as the capacity of a system to absorb disturbances and reorganize to maintain essentially the same functions, structures, properties, and feedbacks in the face of external shocks (Walker et al., 2004). Because of the multidimensional nature of resilience notion, it has been adopted in different scientific disciplines, including physics, risk management, social sciences, and regional economics (Cowell, 2013). Existing studies have mainly focused on evaluating regional resilience (Bruneckiene et al., 2019; Bec et al., 2019) and describing its role in economic development (Cowell, 2013; Di Caro, 2015). However, less literature has drawn attention to the evaluation of regional innovation ecosystem resilience. In addition, there is no clear definition of regional innovation ecosystem resilience and no accepted method on how to measure it.

This study, therefore, attempts to fill these gaps by validating the mediating effect of regional innovation ecosystem resilience on the relationship between government support and digital economy development. If the mediating effect of regional innovation ecosystem resilience can be empirically verified, it is theoretically possible to answer the hotly debated question of how government support can contribute to digital economy development. To examine this question, this paper uses data on 31 provinces in China from 2011 to 2020. Government support, regional innovation system resilience and digital economy development are included in a unified analytical framework to clarify the complex logical relationship among them. This study provides useful policy ideas and empirical evidence for local governments to promote digital economy development.

The main contributions of this paper are fourfold. First, based on internal and external comprehensive consideration, the regional innovation ecosystem resilience evaluation system is constructed. This can accurately assess the ability of the regional innovation ecosystem to withstand external shocks. Second, this paper finds that regional innovation ecosystem resilience plays a full mediating role in the relationship between government support and digital economy development. This contributes to our understanding of the influence mechanism of the two aspects and analyze from a new perspective, and further enriches the existing research on government support and digital economy development. Third, this paper is one of the first efforts to examine the relationship between government support and digital economy development. Existing research on the digital economy primarily focuses on the perspective of measurement and impact effects, and much less is known about its antecedent. Finally, this paper reveals the spatial spillover effects of regional innovation ecosystem resilience on digital economy development. It is rare to fully consider the spatial effects between them in existing research. This is a breakthrough that will be greatly enriched the conclusions of this study.

The paper is organized as follows. The second section provides the research background and theoretical analysis. The third section introduces the data, discusses the construction of the variables of interest and describes the empirical model. The fourth section offers the main findings. The fifth section presents the results of the extended analysis. In the last section, the main conclusions and implications of this study are provided.

2. Theoretical analysis and research hypothesis

2.1. Government support and digital economy development

According to the basic principles of Marxism, the economic base determines the superstructure, and the superstructure has an inverse effect on the economic base (Wang and Guo, 2022). In all areas of the superstructure, politics and its form of realization have the most direct and powerful reaction to the economic base, relative to culture and ideology. To a large extent, political conditions can even influence and determine the path of economic development. The digital economy refers to the use of digital tools and advanced technologies in normal business, such as mobile applications, social networks and e-commerce. Government support mainly refers to the policy support for a new industry or a national strategic industry through special government investment. As a new state and model, the development of the digital economy is closely related to government support. Both scientific research and anecdotal evidence suggest that government policy has a significant impact on the rate and direction of technological progress (Winthrop et al., 2002). Hence, an insightful analysis is required to explore the relationship of government support and digital economy development.

In China, government support for the digital economy industry continues to expand, including investment in science and technology, and R&D subsidies, to improve the level of digital economy development. Government support not only alleviates financial pressure and reduces production costs, but also helps to restructure the industry. At a theoretical level, recent research has examined the impact of government policy on digital economy development. For instance, based on the development status of China's digital economy, Feng and Zhu (2013) and Zhong et al. (2017) analyzed the digital economy development strategies of developed countries such as the US and UK, as well as the new trends in digitization, servitization, intermediation and customization of global investment. They argued that the government should actively play a supporting and exemplary role in promoting the development of the country's digital economy by increasing consumer subsidies and expanding the market demand for the digital economy. Fan and Xu (2020) believed that the fiscal and tax policies were the driving force and basic guarantee of the high-quality development of the digital economy. The main mechanism of financial support for the high-quality development of the digital economy was reflected in the consumption, production, market building, and industrial development ecology.

In addition, several authors have explored the impact of government policies on technological innovation in digital economy industries from an empirical perspective. For instance, Deng et al. (2021) found that R&D subsidies had a significant role in promoting technological innovation in digital enterprises, using data of listed companies in China's digital economy industry in the period 2008–2017. Also, R&D subsidies were found to have a more significant role in promoting R&D investment in the media and Internet service industry. Yu et al. (2021) analyzed the influence of industrial policy on technological innovation in the digital economy industry, including government subsidies, tax incentives, credit facilities, and industry access system. The results showed that government subsidies and industry access system had a significant impact on patent application and invention in digital economy industry. In contrast, tax incentives and credit facilities have a relatively little impact on technological innovation in digital economy industry.

Based on the previous discussions and related literature, we develop the first hypothesis:

Hypothesis 1. Government support is conducive to the improvement of digital economy development.

2.2. Mediating effect of regional innovation ecosystem resilience on government support and digital economy development

Under uncertain environment, enhancing the ability of the regional innovation ecosystem to withstand external shocks will help the region adapt to external changes and achieve high-quality development. The majority of studies have focused on regional resilience, while the resilience of the regional innovation ecosystem is still in its infancy. Thus, there is no universally agreed definition of regional innovation ecosystem resilience. Yang et al. (2021) defined 'regional digital innovation ecosystem resilience' as the ability to recover and adapt from external shocks and pressures. Liang et al. (2020) defined 'national new district innovation ecosystem resilience' as the ability to recover quickly from external shocks to the initial or higher level of functioning. Based on the existing research, we define 'regional innovation ecosystem resilience' as the ability of a region to quickly recover to the initial or higher level of functioning through self-learning and self-adaptation in the face of external shocks and disturbances.

Regional innovation ecosystem resilience is an important way for government support to promote digital economy development. On the one hand, the extant literature has extensively covered the impact of government support on regional innovation performance and region-level resilience. Research into the exact relationship of government support and regional innovation performance has yielded three distinct outcomes, namely positive, negative and convex relationships. Firstly, scholars, including Szczygielski et al. (2017) and Cui and Zhu (2019), put forward the proposition that there was a positive relationship between government support and regional innovation performance. Secondly, scholars, including Zhang (1997) and Yang (2016), argued that government support might have a negative impact on performance. Finally, Wu (2016) found that there was a convex relationship between technology support and regional innovation.

Several authors have explored the impact of government on region-level resilience. For instance, Arbolino and Di Caro (2021) documented that the absorption and allocation of the EU funds could sustain regional labor market resilience by reducing employment losses during COVID-19. Rios and Gianmoena (2020) found that improving the quality of government could help to improve regional responsiveness to economic downturns. Psycharis et al. (2021) maintained that decentralized public investment spending implemented through local governments, as well as spending related to the secondary and tourism sectors, had a positive and significant impact on regional resilience during the economic crisis.

Although a large number of studies have focused on the regional level, these studies are still limited to adequately define the specific role of government on regional innovation performance and regional resilience. In addition, few studies have focused on regional innovation ecosystem resilience. Notably, in a new normal of uncertainty, it is essential to examine the specific influence of regional innovation ecosystem resilience. Accordingly, this study attempts to draw on both literatures to illustrate the relationship of government support and regional innovation ecosystem resilience.

Since the regional innovation ecosystem is faced with high uncertainty and risk, according to the resource-based theory, local governments need to master some heterogeneous and inimitable resources to support innovation activities. The impact of government support on regional innovation ecosystem resilience is reflected in diversity, cushioning, fluidity and evolution. First, government support promotes the diversity of innovation subjects. As an important external resource, government support can supplement the internal innovation resources of innovation subjects (Guo et al., 2016). In addition, it can also attract

other external institutions to cooperate with innovation subjects. These will help to improve innovation efficiency and enhance the adaptability to the external environment. Second, government support provides rich resources for innovation subjects to carry out innovation activities. These resources, including economic, social and technological resources, can mitigate the impact of external shocks on the regional innovative ecosystem. Third, government support can accelerate the flow efficiency of innovative elements. The construction of hard environment such as transportation and information technology, as the leader of regional innovation environment construction, is conducive to the free flow of innovation elements. Moreover, it also accelerates the spread of knowledge and technology in the region, and makes the allocation of innovation resources more reasonable, which provides strong support for the regional innovation ecosystem to withstand external shocks. Finally, government support can improve innovation efficiency. The full support of the government can stimulate the initiative of innovators. Moreover, with the support of government policies, it is beneficial to improve the probability of regional innovation R&D and achievement transformation, improve the level of innovation output, and then enhance regional innovation ecosystem resilience.

On the other hand, regional innovation ecosystem resilience can effectively promote the development of the digital economy. The existing literature has mainly focused on the impact of region-level resilience on digital economy development. For instance, Sun and Yuan (2020) concluded that the development of the digital economy played a mediating role in between economic resilience and high-quality economic development. Zhang and Feng (2019) found that the coordination of urban resilience and economic development in the Yangtze River Delta showed a steady upward trend on the whole. However, the absolute value was still relatively small, and there was a clear differentiation between cities, forming the spatial distribution pattern of "strong in the east and weak in the west." Tang et al. (2021) argued that a resilient region would contribute to its sustainable development. Hu and Yu (2022) believed that innovative resilience made a positive contribute to creative output. However, there is little literature on the influence of regional innovation ecosystem resilience on digital economy development. This paper therefore attempts to bridge this gap. The regional innovation ecosystem has the ability to cope with external shocks and disturbances, which not only helps the region adapt to external changes, but also enhances the development level of the digital economy.

Based on the previous discussion, **Hypothesis 2** is presented as follows:

Hypothesis 2. Regional innovation ecosystem resilience mediates the relationship between government support and digital economy development.

2.3. Spatial spillover effects of regional innovation ecosystem resilience on digital economy development

According to Tobler's law (Tobler, 1970), all things are related to other things, but things closer together are more closely related than things farther away. Some studies have suggested that there are a series of diffusion mechanisms in adopting of local and regional policies, through which similar decisions are made by regions in close proximity (Shipan and Volden, 2008). Other studies have also highlighted the importance of introducing spatial interactions between geographic regions to capture the impact of common regional characteristics on the existence of spillover effect (Haining and Haining, 2003). Existing studies have explored the spatial spillover effects of regional resilience. Fang et al. (2022) found that there was a strong spatial correlation between urban resilience and tourism economy development, and urban resilience had a significant spatial spillover effect on tourism economy development. Simmie (2014) found that the higher resilience in the southeast of UK was due to the agglomeration effect of economic

development in London, and that there was a spatial spillover effect of economic resilience. Therefore, the enhancement of regional resilience is beneficial to the development of regional digital economy. It can also drive the development of the digital economy in the surrounding areas by promoting the inter-regional flow of innovation elements, optimizing the good allocation of innovation resources in the regional space, and accelerating the formation of regional innovation network.

Based on the previous discussion, **Hypothesis 3** is proposed as follows:

Hypothesis 3. Regional innovation ecosystem resilience has a spatial spillover effect on digital economy development.

Our conceptual model and hypotheses are shown in Fig. 1.

3. Research methodology

3.1. Data description

The empirical analysis was carried out on a panel data set of the 31 Chinese provinces (except for Hong Kong, Macao and Taiwan) for the period 2011–2020. Notably, there are two main reasons why Hong Kong, Macao and Taiwan are not included in the sample in this paper: First, due to limited data availability, the data of Hong Kong, Macao and Taiwan have many missing values in the statistics, which is difficult to match with the data of other provinces in China. Second, due to the specificity of their own institutional systems, Hong Kong, Macao and Taiwan are quite different from other regions of China in terms of the operation of the market economy. The inclusion of these regions in the analysis may have an impact on the authenticity and reliability of the research findings, so they are excluded from this paper. In addition, the reasons for taking data from 2011 to 2020 are as follows: on the one hand, the rapid expansion of the scale of China's digital economy and the rapid development of digital technology mainly occurred after 2010 (Yuan et al., 2021). The period 2011–2020 is a critical period for China's digital economy from planning to the high-speed development. On the other hand, the specification of relevant statistics mainly started in 2011.

The data were obtained from the China Statistical Yearbook, China Population and Employment Statistics Yearbook, and China Finance Yearbook. In particular, the regional innovation ecosystem resilience index (college diversity) was provided by the China Higher Education Student Information Network. Besides, it is worth mentioning that the China Population and Employment Statistics Yearbook and China Finance Yearbook are updated to 2019. Therefore, the data for 2020 (level of financial development and diversity of talent) is calculated from the average of 2018 and 2019.

Our main variables of interest are government support, regional innovation ecosystem resilience and digital economy development. The following subsections explain all the variables.

3.2. Explanation of variables

3.2.1. Explained variable

In terms of digital economy development, referring to the research ideas of Zhao et al. (2020), digital economy development was measured from the two aspects of Internet development and digital finance, and

the TOPSIS method was used for calculation. The Internet development index was represented by Internet penetration, Internet related employees, Internet related outputs, and the number of mobile Internet users. The index of digital finance was derived from “Peking University Digital Financial Inclusion Index of China” (Guo et al., 2020). The measurement system of digital economy development in China was constructed following the above five indicators.

3.2.2. Core explanatory variable

Several scholars measured government support by directly providing subsidies for financial and scientific research project, while these support methods needed to be achieved through certain financial expenditures (Wei and Liu, 2015; Li and Yang, 2021). Thus, in this paper, government support was measured by the ratio of government expenditure on science and technology to local fiscal expenditure.

3.2.3. Mediating variable

As regards regional innovation ecosystem resilience, its measurement is still unclear in the existing literature. Liang et al. (2020) selected four dimensions to evaluate the national new district innovation ecosystem resilience, including diversity, fluidity, cushioning, and evolution. This indicator system had better comprehension and operability. Given this, based on Liang et al. (2020), a comprehensive regional innovation ecosystem index was established from the above four dimensions. Among them, diversity reflected the diversity of subjects in the regional innovation ecosystem. Evolution measured the input-output capacity of regional innovation ecosystem. Fluidity measured the flow level of resources in the regional innovation ecosystem. Cushioning reflected the ability of the regional economic and social environment to support the regional innovation ecosystem to withstand shocks. The comprehensive regional innovation ecosystem resilience index was then constructed, as shown in Table 1. Due to the subjective assignment of average weights to each indicator, it was difficult to reflect the actual level of regional innovation ecosystem resilience index. Therefore, TOPSIS was used to calculate the composite index in this study.

3.2.4. Control variables

With reference to Wang et al. (2018), Liu et al. (2020), and Zhong and Mao (2020), the following control variables were added to the regression model: (1) resident wage level (Rwage), measured by the ratio of average wages of non-private sector employees to total population in urban areas; (2) urbanization rate (Urb), measured by the ratio of urban resident population to total population in a region; (3) level of financial development (Df), measured by the ratio of the balance of deposits and loans to GDP; and (4) foreign trade (Ft), measured by the ratio of total imports and exports to GDP.

The descriptive statistics of all variables are shown in Table 2.

3.3. Model construction

3.3.1. Direct effect model

To test the direct effect of government support on digital economy development, the following regression model was used:

$$Dig_{it} = \beta_0 + \beta_1 Gs_{it} + \varphi Z + \mu_i + \delta_t + \varepsilon_{it} \quad (1)$$

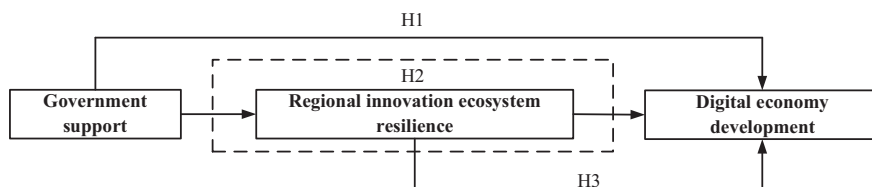


Fig. 1. Conceptual model.

Table 1
Indicator system for measuring regional innovation ecosystem resilience index.

First-level indicators	Second-level indicators	Third-level indicators	Indicators attribute
Evolution	Innovation resources input	R&D personnel full-time equivalent	Positive
		New product development expenses	Positive
	Innovation resources output	Number of domestic patent applications and grants	Positive
Cushioning	Technical resources	New product sales revenue	Positive
		Invention patent ownership per 10,000 people	Positive
	Economic resources	Per capita GDP	Positive
	Natural environment resources	Per capita water resources	Positive
		Exhaust gas per unit output	Negative
Fluidity	Social environment resources	Number of books in public libraries per 10,000 people	Positive
		Number of beds in medical and health institutions per 10,000 people	Positive
	Cash flow	Fixed-asset investment	Positive
Diversity	Freight flow	Rail freight flow	Positive
		Road freight flow	Positive
	Diversity of talent	The proportion of employed personnel with a college degree	Positive
		The proportion of employed persons with a bachelor's degree	Positive
		The proportion of the employed with a post-graduate degree	Positive
		The proportion of higher vocational colleges in the number of colleges and universities	Positive
	College diversity	The proportion of ordinary undergraduates in the number of colleges and universities	Positive
		The proportion of key institutions (985 and 211) in higher education institutions	Positive

where Dig_{it} represents the level of digital economy development; Gs_{it} represents government support; Z represents a series of control variables; i represents the province; t represents the year; β_0 represents the intercept term; μ_i represents province fixed effect; δ_t represents time fixed effect, and ε_{it} represents a random error term.

3.3.2. Mediating effect model

To examine whether government support can promote digital economy development by improving regional innovation ecosystem resilience, with reference to Baron and Kenny (1986), this paper utilized a three-step method to build a mediating effect model, as follows:

$$Dig_{it} = C + \partial Gs_{it} + \varphi Z + \mu_i + \delta_t + \varepsilon_{it} \quad (2)$$

$$Rier_{it} = C + \eta Gs_{it} + \varphi Z + \mu_i + \delta_t + \varepsilon_{it} \quad (3)$$

Table 2
Variable description.

Variable character	Variable name	Variable symbol	Obs.	Mean	Std. Dev.	Min	Max
Explained variable	Digital economy development	Dig	310	0.371	0.174	0.773	0.982
Core explanatory variable	Government support	Gs	310	0.020	0.145	0.003	0.068
Mediating variable	Regional innovation ecosystem resilience	Rier	310	0.168	0.100	0.053	0.451
Control variables	Resident wage level	Rwage	310	0.003	0.005	0.000	0.032
	Urbanization rate	Urb	310	0.579	0.133	0.228	0.938
	Level of financial development	Df	310	3.378	1.102	1.688	7.552
	Foreign trade	Ft	310	0.109	0.164	0.000	0.984

$$Dig_{it} = C + \theta Gs_{it} + \lambda Rier_{it} + \varphi Z + \mu_i + \delta_t + \varepsilon_{it} \quad (4)$$

Where $Rier_{it}$ represents the mediating variable. Eq. (2) measures the impact of government support on digital economy development, Eq. (3) describes the impact of government support on regional innovation ecosystem resilience, and Eq. (4) indicates the mediation impact of regional innovation ecosystem resilience on the relationship between government support and digital economy development. If the coefficient ∂ in Eq. (2) is significant, and both η in Eq. (3) and λ in Eq. (4) are significant, the existence of a mediating effect is confirmed; If the coefficient θ in Eq. (4) is significant, there is a partial mediating effect; if θ is not significant, there is a complete mediation effect.

4. Empirical results

4.1. Baseline regression results

Table 3 presents the baseline regression results of our panel dataset. The model includes the province fixed effects to control for the unobserved and time-invariant heterogeneity across provinces, as well as the year fixed effects to control for unobserved cycles, mitigating the omitted variable problem. Standard errors of point estimates are clustered at the province level to allow for arbitrary within-province autocorrelation. The R^2 values of all the regression models are around 0.9, indicating that the goodness of fit of these models is reassuring. All variables, except for the control variables, are in their logarithmic forms to reduce the impact of heteroscedasticity.

In column (1) of Table 3, we only examine the impact of all the control variables on digital economy development in Chinese provinces. The sign of the coefficients of our control variables remains consistent across all specifications, although there are slight differences in their magnitudes and statistical significance. Overall, the result suggests that a province's digital economy development is positively associated with the level of its resident wage level, urbanization rate, and foreign trade. This is generally in line with the findings of other studies (Liu et al., 2020; Yu, 2021). The coefficient on the level of financial development has a positive sign but is not statistically significant, which may be related to the different levels of financial development in the East and West. Therefore, the coefficient of the level of financial development on the digital economy is not significant.

In column (2) of Table 3, we investigate the impact of government support on digital economy development without considering the effect of other control variables. The coefficient of government support is positive and statistically significant at the 10 % level, suggesting that government support in a province of China is positively associated with its digital economy development.

In columns (3) of Table 3, we include both the control variables and the indicator of government support in the regression model. The indicator of government support is statistically significant, and its coefficient sign remains unchanged compared with those in columns (1) and (2). The result further indicates that government support can improve digital economy development.

Table 3
Results of baseline regression analysis.

Variables	(1)	(2)	(3)
	lnDig	lnDig	lnDig
Rwage	6.2513** (2.6479)		7.3984*** (2.6042)
Urb	1.9475*** (0.6624)		1.7306** (0.6354)
Ft	0.6754*** (0.2191)		0.7260*** (0.2068)
Df	0.0000 (0.0000)		0.0000 (0.0000)
lnGs		0.0769* (0.0431)	0.0512* (0.0254)
Constant	−2.3356*** (0.3693)	−0.7976*** (0.1778)	−2.0083*** (0.3965)
Observations	310	310	310
R-squared	0.9903	0.9811	0.9906
Province FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: All variables, except for control variables, are logarithm transformed. Robust standard errors shown in parentheses are clustered at the province level. ***p < 0.01, **p < 0.05, *p < 0.1. FE, fixed effects.

4.2. Robustness test

Table 4 presents the results of robustness check. In columns (1) and (2) of Table 4, we check the robustness of the baseline regression results with one and two lags of the independent variable, respectively. In addition, in columns (3) and (4) of Table 4, we also use the sample estimate intervals of 2011–2019 and 2012–2020 for robustness check. It is reassuring to see four specifications show that the coefficients of government support are positive and statistically significant at the 10 %, 5 %, 10 %, and 10 % levels, respectively. The robustness check further confirms that government support significantly promotes digital economy development.

4.3. Endogenous test

The baseline regression results provided preliminary validation of the promoting effect of government support on digital economy development. To ensure the reliability of the baseline regression, we used the IV method to address the endogeneity issues, which could cause by

Table 4
Results of robustness check.

Variables	(1)	(2)	(3)	(4)
	lnDig	lnDig	lnDig	lnDig
L.lnGs	0.0384* (0.0203)			
Rwage	5.8423* (3.3543)	6.5430* (3.2148)	5.6239** (2.6555)	5.7687 (3.5611)
Urb	1.4145** (0.5198)	1.1096** (0.4430)	1.9132** (0.7283)	1.4350** (0.5247)
Ft	0.5090*** (0.1693)	0.4421*** (0.1342)	0.7129*** (0.2235)	0.5114*** (0.1708)
Df	−0.0001*** (0.0000)	−0.0002*** (0.0000)	0.0001** (0.0000)	−0.0001*** (0.0000)
L2.lnGs		0.0541** (0.0224)		
lnGs			0.0490* (0.0261)	0.0359* (0.0203)
Constant	−1.7462*** (0.3200)	−1.4207*** (0.2853)	−2.1809*** (0.4473)	−1.7694*** (0.3284)
Observations	279	248	279	279
R-squared	0.9925	0.9927	0.9901	0.9925
Province FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: Robust standard errors shown in parentheses are clustered at the province level. ***p < 0.01, **p < 0.05, *p < 0.1. FE, fixed effects.

reverse causality, omitted variables, and measurement errors. With reference to previous studies, one and two lags of the independent variable were used as instrumental variables. The two instrumental variables were named IV1 and IV2, respectively. The two-stage least squares (2SLS) method was used to conduct endogeneity test. The results of the IV estimation are reported in Table 5.

In columns (1) and (3) of Table 5, we report the estimation results of the first-stage regression using the IV1 and IV2. It can be seen that the coefficients of the two instrumental variables are positive and statistically significant at the 1 % level. The results show that there is a strong positive correlation between instrumental variables and government support. In columns (2) and (4) of Table 5, we report the estimation results of the two-stage regression. The coefficients of government support are 0.0542 and 0.1265, respectively, and they are statistically significant at the 10 % and 5 % levels, respectively. All specifications pass the under-identification test and the weak identification test. The IV results show that after considering the possible endogeneity issues, government support still has a significant positive role in promoting digital economy development, which further ensures the robustness of the baseline regression results.

4.4. Mechanism analysis

Table 6 presents the empirical results of mediating effect. In column (1) of Table 6, we report the total effect of government support on digital economy development. The estimated coefficient is positive and statistically significant at the 10 % level; in other words, government support can effectively promote digital economy development in China. In column (2) of Table 6, we investigate the impact of government support on regional innovation ecosystem resilience. The estimated coefficient is positive and statistically significant at the 1 % level, which shows that

Table 5
Two-stage least squares (2SLS) estimation results.

Dependent variable:	(1)	(2)	(3)	(4)
	lnGs	lnDig	lnGs	lnDig
lnGs		0.0542* (0.0280)		0.1265** (0.0583)
IV1	0.7080*** (0.0594)			
IV2			0.4279*** (0.0942)	
Rwage	−7.0089 (6.9464)	6.2221 (3.6811)	−3.1622 (12.9201)	6.9429 (4.5216)
Urb	0.9572 (0.7561)	1.3627** (0.5064)	1.9240* (1.1283)	0.8663** (0.4198)
Ft	−0.4307 (0.2671)	0.5323*** (0.1654)	−1.1055** (0.4731)	0.5819*** (0.1406)
Df	−0.0003* (0.0002)	−0.0001*** (0.0000)	0.0003 (0.0002)	−0.0002*** (0.0000)
Constant	−1.6707*** (0.5796)		−3.3371*** (0.8885)	
Under-identification test		15.280***		8.787***
Kleibergen–Paap rk LM statistic				
Weak identification test		253.075		48.563
Cragg–Donald Wald F-statistic				
Stock–Yogo critical value (10 % maximal IV)		16.38		16.38
Province FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	279	279	248	248
R-squared	0.9643	0.4975	0.9484	0.2831

Notes: Robust standard errors shown in parentheses are clustered at the province level. ***p < 0.01, **p < 0.05, *p < 0.1. FE, fixed effects.

Table 6
Mechanism verification.

Variables	(1)	(2)	(3)
	lnDig	lnRier	lnDig
lnGs	0.0512* (0.0254)	0.1292*** (0.0364)	0.0240 (0.0226)
lnRier			0.2102** (0.0881)
Rwage	7.3984*** (2.6042)	−10.1337*** (2.6794)	9.5283*** (2.7073)
Urb	1.7306** (0.6354)	0.8303** (0.3878)	1.5561** (0.5913)
Ft	0.7260*** (0.2068)	0.0835 (0.2075)	0.7085*** (0.1871)
Df	0.0000 (0.0000)	−0.0002*** (0.0001)	0.0001** (0.0000)
Constant	−2.0083*** (0.3965)	−1.8582*** (0.2605)	−1.6177*** (0.3995)
Observations	310	310	310
R-squared	0.9906	0.9883	0.9911
Province FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: All variables, except for control variables, are logarithm transformed. Robust standard errors shown in parentheses are clustered at the province level. ***p < 0.01, **p < 0.05, *p < 0.1. FE, fixed effects.

government support can significantly promote regional innovation ecosystem resilience. In columns (3) of Table 6, we investigate the impact of government support on digital economy development by promoting regional innovation ecosystem resilience. The estimated coefficient of government support is insignificant, while the estimated coefficient of regional innovation ecosystem resilience is positive and statistically significant at the 5 % level. This indicates that regional innovation ecosystem resilience plays a full mediating role between government support and digital economy development; that is, government support enhances digital economy development by promoting regional innovation ecosystem resilience. This possible explanation is that the rapid development of China's digital economy cannot be separated from the good construction of the regional innovation ecosystem. In the current uncertain environment, the government as an important part of regional innovation system, government support is conducive to enhance the ability of regional innovation ecosystem to withstand external shocks, and then promote digital economy development.

5. Extended analysis

5.1. Spatial Durbin model

According to hypothesis 3, the spatial Durbin model (SDM) was established to verify the spillover effect of regional innovation ecosystem resilience on digital economy development. The spatial Durbin model was constructed as follows:

$$Dig_{it} = C + \rho W Dig_{it} + \psi_1 W Rier_{it} + \lambda Rier_{it} + \psi_2 W Z + \varphi Z + \mu_i + \delta_t + \varepsilon_{it} \quad (6)$$

In the model, W represents the geographic distance spatial weight matrix. W is built through three steps: firstly, the distance between provinces is calculated according to the latitude and longitude of the provincial center; secondly, the reciprocal of the distance is taken as the element of the spatial weight matrix; finally, the spatial weight matrix is standardized to calculate the W used in the model. ρ represents the spatial autoregressive coefficient.

5.2. Spatial autocorrelation test

Before the spatial econometric analysis, it is necessary to test whether spatial autocorrelation exists between regional innovation ecosystem resilience and digital economy development. The Moran's I of 2011–2020 is shown in Table 7. It can be seen from Table 7 that regional

innovation ecosystem resilience and digital economy development have a significant positive spatial correlation.

5.3. Spatial spillover effects analysis

Table 8 presents the decomposition results of the spatial panel Durbin model. In column (1) of Table 8, we investigate the direct effect of regional innovation ecosystem resilience on digital economy development. The estimated coefficient for regional innovation ecosystem resilience is positive and statistically significant at the 1 % level. This shows that regional innovation ecosystem resilience can promote digital economy development in China. The level of digital economy development will significantly increase by 0.2694 for every unit increase in regional innovation ecosystem resilience. In column (2) of Table 8, we examine the indirect effect of regional innovation ecosystem resilience on digital economy development. The coefficient is positive and statistically significant at the 5 % level, which indicates that the impact of regional innovation ecosystem resilience on digital economy development has a significant spatial spillover. The level of digital economy development of neighboring provinces will significantly increase by 3.4228 for every unit increase in regional innovation ecosystem resilience of the local provinces. In column (3) of Table 8, we explore the total effect of regional innovation ecosystem resilience on digital economy development. The coefficient is positive and statistically significant at the 5 % level, which suggests that regional innovation ecosystem resilience affects the level of digital economy development through direct or indirect effects. In general, the improvement of regional innovation ecosystem resilience can not only strengthen the local digital economy development, but also generate a strong spatial spillover effect on neighboring regions.

6. Conclusions and implications

Based on the construction of a comprehensive index system of regional innovation ecosystem resilience, this study examines the impact of government support on digital economy development. It also investigates the mediating role of regional innovation ecosystem resilience between government support and digital economy development. Furthermore, the spatial spillover effects of regional innovation ecosystem resilience on digital economy development are analyzed. Using a panel data covering 31 provinces of China from 2011 to 2020, the main conclusions are presented as follows:

(1) The results of the baseline regression indicate that government support promotes digital economy development. The results are consistent with other studies in the Chinese context (Fan and Xu, 2020; Deng et al., 2021).

(2) The empirical results of the mediating effect model show that regional innovation ecosystem resilience plays a full mediating role between government support and digital economy development; in other words, government support promotes digital economy

Table 7Moran's I test.

Year	Regional innovation ecosystem resilience			Digital economy development		
	Moran's I	Z-value	p-value	Moran's I	Z-value	p-value
2011	0.082	3.294	0.000	0.073	3.097	0.001
2012	0.086	3.421	0.000	0.080	3.359	0.000
2013	0.084	3.346	0.000	0.071	3.107	0.001
2014	0.085	3.375	0.000	0.063	2.937	0.002
2015	0.086	3.405	0.000	0.047	2.450	0.007
2016	0.081	3.286	0.001	0.062	2.944	0.002
2017	0.068	2.917	0.002	0.044	2.379	0.009
2018	0.070	2.961	0.002	0.032	1.984	0.024
2019	0.075	3.108	0.001	0.029	1.898	0.029
2020	0.086	3.424	0.000	0.027	1.861	0.031

Table 8

Spatial panel Durbin model effect decomposition results.

Variables	(1)	(2)	(3)
	Direct effect	Indirect effect	Total effect
lnRier	0.2694*** (0.0978)	3.4228** (1.6187)	3.6922** (1.6650)
Rwage	6.3654 (11.7885)	−0.6718 (341.9306)	5.6936 (353.4231)
Urb	1.8765*** (0.5356)	13.2091* (6.8867)	15.0856** (7.0670)
Ft	0.9707*** (0.3353)	10.0738 (9.3563)	11.0445 (9.6582)
Df	−0.0007 (0.0005)	−0.0219 (0.0140)	−0.0226 (0.0145)
Observations	310	310	310
R-squared	0.8033	0.8033	0.8033
Number of provinces	31	31	31
Province FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: Robust standard errors shown in parentheses are clustered at the province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. FE, fixed effects.

development through the mediating mechanism of regional innovation ecosystem resilience.

(3) The empirical results of the spatial Durbin model show that regional innovation ecosystem resilience has a significant positive spillover effect on digital economy development; in other words, regional innovation ecosystem resilience not only promotes digital economy development in this region, but also effectively promotes digital economy development in neighboring regions. The regional innovation ecosystem resilience shows the characteristics of “being good to neighbors” in space.

Based on the aforementioned findings, the study has the following implications. First, considering the role of government support in promoting digital economy development, it not only expands the antecedents of digital economy development, but also provides guidance for how to promote it. It is necessary for all regions to seize opportunities and turn policy dividends into the driving force for digital economy development. On the one hand, it is important to make full use of the basic industrial policy, talent policy, and land policy provided by the government. These government policies have laid a substantial foundation for digital economy development, thus contributing to making the digital economy bigger and stronger. On the other hand, government support for digital economy development should be further strengthened. The government should give full play to the guiding role of financial funds, and invest more financial funds in scientific and technological innovation and the digital economy industry to promote digital economy development.

Second, the study finds that regional innovation ecosystem resilience has a full mediating role between government support and digital economy development. This result suggests that regional innovation ecosystem resilience is a vital transmission path for government support to promote digital economy development. Thus, the government should focus on improving regional innovation ecosystem resilience in order to effectively promote digital economy development. On the one hand, the government should increase investment in scientific and technological innovation, fully mobilize and integrate social innovation resources, accelerate the gathering of innovative talents, and strengthen the communication between innovation subjects. On the other hand, the government should create a good regional innovation environment, improve innovation efficiency, and accelerate the transformation of innovation results.

Third, the spatial spillover effects of regional innovation ecosystem resilience on digital economy development are considered. This result provides theoretical support for how backward provinces can improve the level of digital economy development by relying on regional innovation ecosystem resilience. To better promote digital economy

development, all areas should take the opportunity to enhance regional innovation ecosystem resilience, actively build regional cooperation platforms, and form a closely coordinated regional innovation chain. In addition, it is necessary to enhance the sharing of public innovation resources and infrastructure, and promote the free flow of innovation elements between regions.

This article has some limitations. First, the study focuses only on region-level to investigate the impact of government support on digital economy development. Future research could use city-level data to examine the link between the two. Second, the study preliminarily discusses the spatial spillover effects of regional innovation ecosystem resilience on digital economy development. It would be interesting to explore whether differences in indicator systems, study regions, and time intervals lead to changes in the relationship between the two. Last, due to the limited availability of data, on the one hand, the study only measures the level of digital economy development from the two aspects of Internet development and digital finance. Without considering other important digital technologies such as artificial intelligence, it may be difficult to accurately summarize the connotation of digital economy. On the other hand, gender diversity at the micro-level has not been considered, which may neglect the value of gender diversity in enhancing regional innovation ecosystem resilience, and may lead to the lack of comprehensive diversity characteristics. These aspects need to be further explored as future research directions.

Credit authorship contribution statement

Lin Liang: Conceptualization, Data curation, Writing – original draft, Writing – review & editing, Supervision, Project administration and funding acquisition. Yan Li: Conceptualization, Data curation and software, Writing – original draft, Writing – review & editing, formal analysis.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Data availability

Data will be made available on request.

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